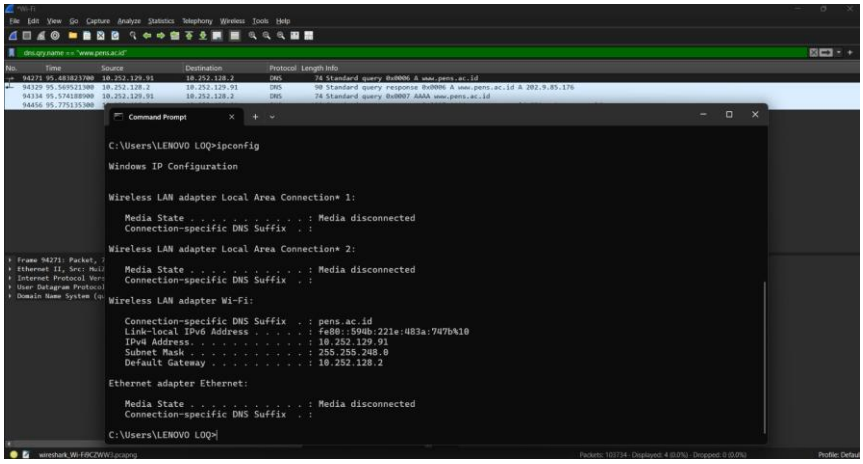
Laporan Praktikum 2. TCP/IP Model

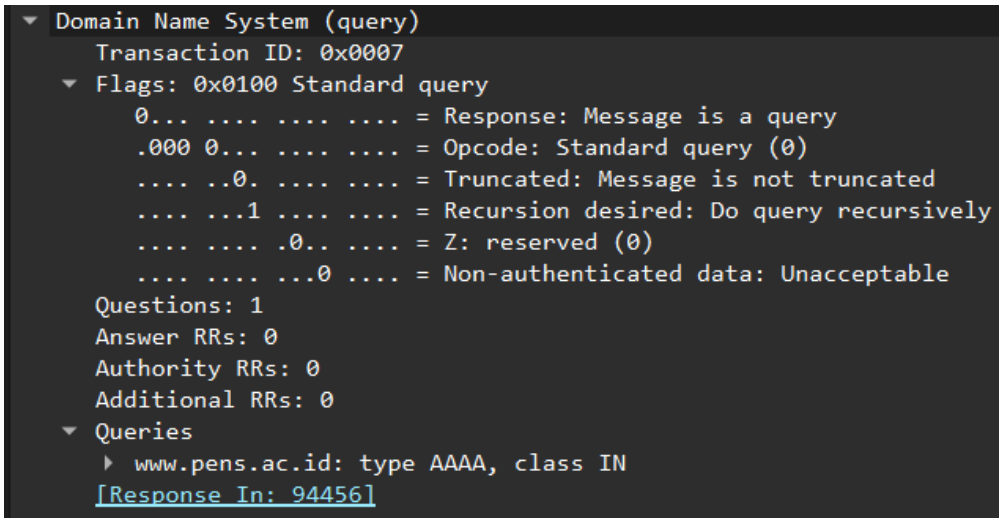
Kegiatan: Identifikasi OSI Layer DNS www.pens.ac.id
Nama: Bian Mulia Rahmatullah
NRP: 5325600006

IP

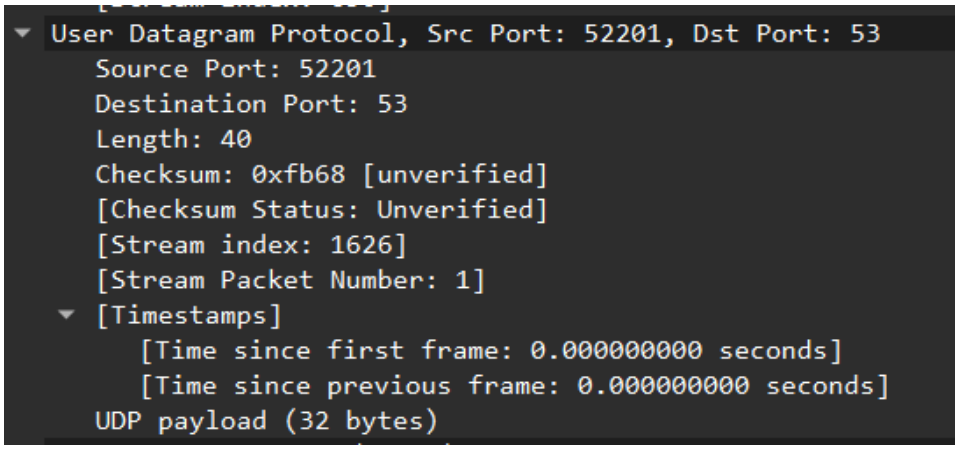


Data Layer

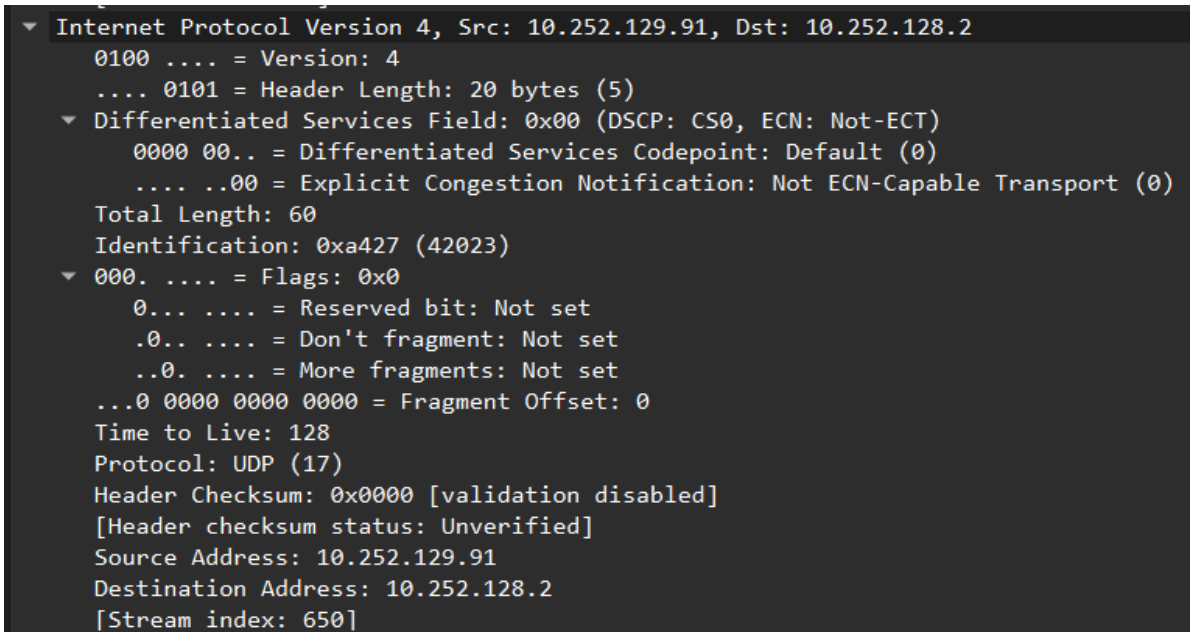
Application layer



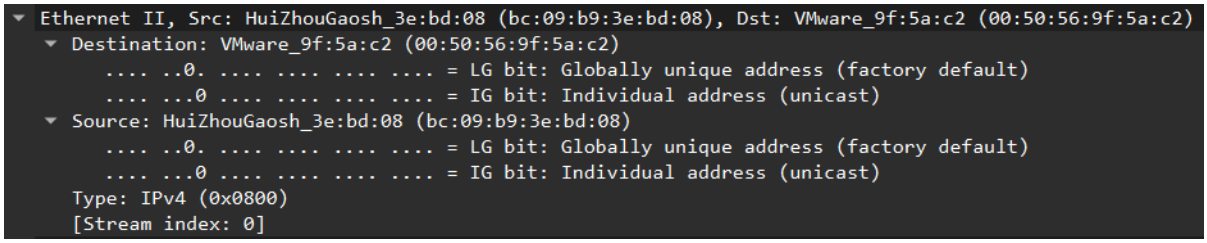
Host-to-host layer



Internetwork layer



Network Interface layer



Laporan Praktikum 2. TCP/IP Model

Kegiatan: Identifikasi OSI Layer FTP newfs.pens.ac.id

Nama: Bian Mulia Rahmatullah

NRP: 5325600006

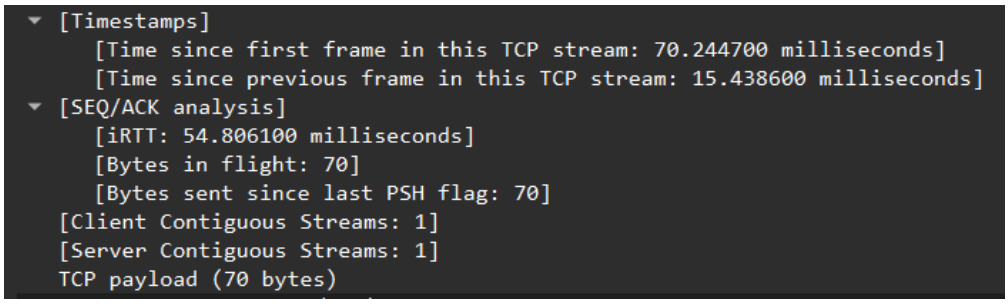
Data Layer

Application layer

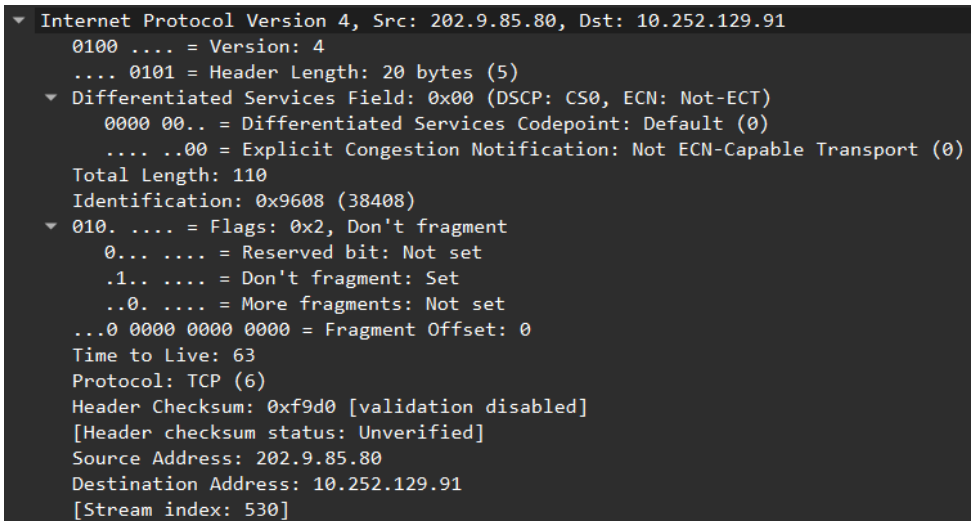
```
File Transfer Protocol (FTP)
  220 ProFTPD 1.3.5b Server (newfs.eepis-its.edu) [::ffff:202.9.85.80]\r\n
      Response code: Service ready for new user (220)
      Response arg: ProFTPD 1.3.5b Server (newfs.eepis-its.edu) [::ffff:202.9.85.80]
  [Current working directory: ]
```

Host-to-host layer

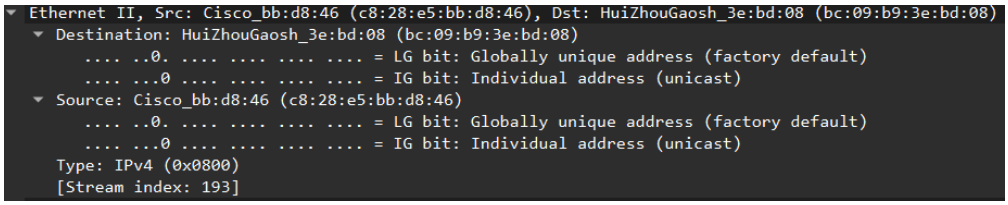
```
Transmission Control Protocol, Src Port: 21, Dst Port: 64281, Seq: 1, Ack: 1, Len: 70
    Source Port: 21
    Destination Port: 64281
    [Stream index: 11]
    [Stream Packet Number: 4]
  [Conversation completeness: Incomplete, DATA (15)]
      ..0. .... = RST: Absent
      ...0 .... = FIN: Absent
      .... 1... = Data: Present
      .... .1.. = ACK: Present
      .... ..1. = SYN-ACK: Present
      .... ...1 = SYN: Present
      [Completeness Flags: ··DASS]
    [TCP Segment Len: 70]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 3716405686
    [Next Sequence Number: 71    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 3013312450
    0101 .... = Header Length: 20 bytes (5)
  Flags: 0x018 (PSH, ACK)
      000. .... .... = Reserved: Not set
      ...0 .... .... = Accurate ECN: Not set
      .... 0... .... = Congestion Window Reduced: Not set
      .... .0.. .... = ECN-Echo: Not set
      .... ..0. .... = Urgent: Not set
      .... ...1 .... = Acknowledgment: Set
      .... .... 1... = Push: Set
      .... .... .0.. = Reset: Not set
      .... .... ..0. = Syn: Not set
      .... .... ...0 = Fin: Not set
      [TCP Flags: ·····AP···]
    Window: 229
    [Calculated window size: 29312]
    [Window size scaling factor: 128]
    Checksum: 0x6787 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
```



Internetwork layer



Network Interface layer



Laporan Praktikum 2. TCP/IP Model

Kegiatan: Identifikasi OSI Layer HTTP http://www.pens.ac.id

Nama: Bian Mulia Rahmatullah

NRP: 5325600006

Data Layer

Application layer

```
▾ Hypertext Transfer Protocol
  ▾ GET / HTTP/1.1\r\n
      Request Method: GET
      Request URI: /
      Request Version: HTTP/1.1
    Host: www.pens.ac.id\r\n
    User-Agent: curl/8.21.0\r\n
    Accept: */*\r\n
    \r\n
    [Response in frame: 347920]
    [Full request URI: http://www.pens.ac.id/]
```

Host-to-host layer

```
▾ Transmission Control Protocol, Src Port: 51701, Dst Port: 80, Seq: 1, Ack: 1, Len: 78
    Source Port: 51701
    Destination Port: 80
    [Stream index: 85]
    [Stream Packet Number: 4]
  ▾ [Conversation completeness: Complete, WITH_DATA (31)]
      ..0. .... = RST: Absent
      ...1 .... = FIN: Present
      .... 1... = Data: Present
      .... .1.. = ACK: Present
      .... ..1. = SYN-ACK: Present
      .... ...1 = SYN: Present
      [Completeness Flags: ·FDASS]
    [TCP Segment Len: 78]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 1862049357
    [Next Sequence Number: 79    (relative sequence number)]
    Acknowledgment Number: 1    (relative ack number)
    Acknowledgment number (raw): 578348589
    0101 .... = Header Length: 20 bytes (5)
  ▾ Flags: 0x018 (PSH, ACK)
      000. .... .... = Reserved: Not set
      ...0 .... .... = Accurate ECN: Not set
      .... 0... .... = Congestion Window Reduced: Not set
      .... .0.. .... = ECN-Echo: Not set
      .... ..0. .... = Urgent: Not set
      .... ...1 .... = Acknowledgment: Set
      .... .... 1... = Push: Set
      .... .... .0.. = Reset: Not set
      .... .... ..0. = Syn: Not set
      .... .... ...0 = Fin: Not set
      [TCP Flags: ·····AP···]
    Window: 255
    [Calculated window size: 65280]
    [Window size scaling factor: 256]
    Checksum: 0xac79 [unverified]
    [Checksum Status: Unverified]
```

```
    Urgent Pointer: 0
  ▾ [Timestamps]
      [Time since first frame in this TCP stream: 29.919500 milliseconds]
      [Time since previous frame in this TCP stream: 17.105700 milliseconds]
  ▾ [SEQ/ACK analysis]
      [iRTT: 12.813800 milliseconds]
      [Bytes in flight: 78]
      [Bytes sent since last PSH flag: 78]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (78 bytes)
```

Internetwork layer

```
▾ Internet Protocol Version 4, Src: 10.252.129.91, Dst: 202.9.85.176
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  ▾ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
      0000 00.. = Differentiated Services Codepoint: Default (0)
      .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 118
    Identification: 0x0bff (3071)
  ▾ 010. .... = Flags: 0x2, Don't fragment
      0... .... = Reserved bit: Not set
      .1.. .... = Don't fragment: Set
      ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.252.129.91
    Destination Address: 202.9.85.176
    [Stream index: 1785]
```

Network Interface layer

```
▾ Ethernet II, Src: HuiZhouGaosh_3e:bd:08 (bc:09:b9:3e:bd:08), Dst: VMware_b5:55:03 (00:50:56:b5:55:03)
  ▾ Destination: VMware_b5:55:03 (00:50:56:b5:55:03)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  ▾ Source: HuiZhouGaosh_3e:bd:08 (bc:09:b9:3e:bd:08)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    [Stream index: 843]
```